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PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: The contribution of non-targeted effects in HZE cancer risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Task Description:	Clear mechanistic understanding of the biological processes elicited by high LET radiation increase cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects like cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness. However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we will develop a comprehensive data base of normal tissue and irradiated tissue by analyzing different organs in each mouse. This data base will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer.
Task Progress:	Logistics The second year of the funding was devoted in part to staffing and training the three projects encompassed in this NASA Specialized Center of Research. An additional postdoctoral fellow, Cassandra Adams, was recruited to work on Project 2, and has completed training for BNL and NSRL access and use. The PI and co-investigators met three times to discuss the research. In the second year, we have completed three experiments at NSRL using three particle beams. We have irradiated. Project 1 We use a novel radiation/genetic mammary chimera in which either the whole mouse, the host microenvironment, or the transplanted epithelium is irradiated. Preliminary data show that NTE dramatically increases tumor development in hosts irradiated prior to being transplanted with p53 null mammary epithelium. This model will be used to evaluate the relative contribution of targeted vs non-targeted radiation effects from low vs high LET and to determine whether the contribution of NTE is impacted by age, or that chronic low dose rate exposure to low LET (i.e. protons) will interact with acute high LET radiation. Aim 1 consists of tumor incidence and latency measured for gamma-rays and two fluences of three ions (600 MeV Fe and 350 MeV Si or Ar) that will require two years to complete for each ion. Two experiments consisting of hosts irradiated with either Fe or Si particles have been initiated during the Fall 09 and Spring 10 runs. The data from the radiation chimera model on which Project 1 is based was published in the May 2011 issue of Cancer Cell, which co-authored by NSCOR members, Drs. I. Illa-Bochaca, Mao and Barcellos-Hoff. We report that mice irradiated with low doses (10-100 cGy) and subsequently orthotopically transplanted with p53 null mammary epithelium undergo accelerated tumorigenesis of predominantly estrogen receptor negative tumors. One of the most notable new findings resulting from this study is that radiation-induced activation of Notch signaling may contribute to altered stem cell self-renewal. Preliminary data suggest that the size of the stem/progenitor compartment is also affected by densely ionizing radiation. Notch and beta-catenin immunolocalization in particle irradiated tissue is being conducted to determine how radiation quality affects the magnitude of this effect. The contribution of inflammation as a mediator of NTE carcinogenic risk studied by Drs. Demaria and Matsumura in Aim 2 using systemic and local functional assays. Overall, data suggest that all radiation regimens cause a mild inflammatory response. The number of total spleen cells was detected in mice treated with gamma or high dose Fe at 1 week was reduced by 50%, with return towards the sham level at 1 month. Microarray expression profiling resonates with the immune function studies in that inflammation associated genes are enriched in the expression profile of mammary gland post-irradiation. Project 2 The goal of this project is to use genetic analysis as an unbiased means to reveal the contribution of different biological components and/or processes. We hypothesize that organ-specific risks have different genetic determinants/associations, and that radiation quality differentially affects the genetic architecture of gene expression in radiosensitive tissues. Expression profiles in tissues that are sensitive to the effects of radiation generate a network view of gene expression. We will investigate how this network is perturbed by genetic manipulation (alterations in the p53 pathway), and by exposure to high or low LET radiation. To facilitate comparisons with earlier gene network analysis, the p53 delta-P mouse is currently being bred onto a pure FVB background for further timepoint and longer term tumorigenesis studies, which commenced in 2011. In addition, we will investigate responses of p53 knockout mice on various genetic backgrounds to high and low LET radiation. Project 3 To provide an integrated mechanistic model of the biology, Monte Carlo models of DNA damage will be used with agent-based models (ABM) to simulate the dynamic response of tissue to radiation. This modeling approach is used to

	evaluate and prioritize key events during radiation-preceded carcinogenesis that result from immediate targeted DNA damage and persistent non-targeted effects. Extended time series will use gene expression motifs identified for radiosensitive tissues in Projects 1 and 2 translated into functions (proliferation, apoptosis) of either the agents or the microenvironment. we have constructed a computational systems model of the mouse mammary gland using an agent-based formalism. Agents are autonomous software objects that have defined rules or mathematical equations that determine how they behave and interact with each other and their local environment. Our in silico mammary gland consists of interacting agents simulating either mammary epithelial cells or stromal cells, in a simulation environment with variables representing extracellular matrix properties. We have modeled normal murine mammary morphogenesis and maintenance by establishing a set of rules and parameters that led to cumulative agent behavior that match experimental data of the developing gland. Our next objective is to model how radiation can perturb each of mammary gland compartments and lead to a disorganized morphogenesis, i.e. dysplasia, a hallmark of cancer. In combination with modeling microdosimetry of various radiation qualities in this system, this tool will allow us to test how geometrical properties of energy deposition can modify relative biological effectiveness. This exercise will help refine hypotheses in which radiation carcinogenesis is promoted by synergistic interaction between targeted damage and non-targeted effects.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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